

KEYUR RAKESHBHAI RANA

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PROFESSIONAL SUMMARY

Hardware and Embedded Systems Engineer specializing in IoT solutions, microcontrollers, PCB integration, and embedded UI design (LVGL/DGUS). Experienced in developing autonomous robotics and full-stack IoT mobile/web applications.

EXPERIENCE

Embedded & IoT Engineer Intern

May 2026 – July 2026

TECHYMONK, Ahmedabad

- Engineered an IR Blaster module and developed a low-cost water temperature measurement device for real-time monitoring.
- Designed and prototyped a multi-planetary weight measurement system using embedded sensor modules.
- Mastered the LVGL graphics library and DGUS software to build interactive HMI screen interfaces for embedded displays.
- Tested, debugged, and integrated custom hardware components on production-ready PCB circuits.

Embedded Systems Intern

June 2024 – August 2024

Porji System and Technology, Ahmedabad

- Programmed microcontrollers using Embedded C and Arduino framework for real-world hardware control applications.
- Interfaced 8051 microprocessors with various electronic components, sensors, and actuators.
- Conducted hardware troubleshooting, schematic validation, and hands-on component testing.

Python Programmer Intern

August 2023 – August 2023

Spyderwave Technology, Ahmedabad

- Developed and optimized 100+ Python scripts focused on core data structures, algorithms, and logical problem-solving.
- Strengthened foundational knowledge in Python programming paradigms and modular code development.

PROJECTS

MediBot: Autonomous Medical Capsule for Hospital Logistics & Safety

Tech Stack: ESP32, C++, ToF LiDAR, React Native, MERN Stack (Node.js, Express, MongoDB), WebSockets

- Developed an autonomous robotic transport system for secure medical logistics using ESP32, 5-channel IR array, and TF-Luna ToF LiDAR for 100Hz real-time obstacle detection.
- Authored deterministic C++ firmware for sub-millisecond emergency braking (≤ 30 cm) and intersection-based navigation.

- Built a full-stack IoT dashboard (MERN & React Native) with WebSockets, Role-Based Access Control (5 roles), and dynamic PDF invoice generation.
- Integrated an electronic deadbolt cargo lock with metal-gear servo actuation and delivered a functional prototype under a budget of 4,590.

IoT-Based Smart Agriculture Monitoring and Control System

Tech Stack: *ESP32, React Native (Expo Go), Python (Pandas, Matplotlib), Relay Module, Sensors (DHT11, MQ-135, TDS)*

- Designed an IoT prototype integrating ESP32 with Soil Moisture, DHT11, MQ-135, LDR, and TDS sensors for real-time environmental tracking.
- Developed a React Native mobile application for real-time sensor visualization and remote automated irrigation control.
- Automated pump operation based on live soil moisture telemetry and conducted unit/integration testing for sensor accuracy.
- Performed Python-based data analysis (Pandas, Matplotlib) to lay the groundwork for predictive ML-based irrigation.

SKILLS

Languages & Domains: Python, C++, C, IoT, Embedded Systems, LVGL Library, DGUS, Embedded C, React Native, MERN Stack

Hardware & Sensors: ESP32, 8051, Microcontrollers, ToF LiDAR, IR Arrays, Relay Modules, Actuators, PCB Testing

Tools & Platforms: Git, GitHub, VS Code, Antigravity, Cursor, Docker, Arduino IDE, Proteus

Soft Skills: Problem Solving, Leadership, Creative Thinking, Brainstorming, Project Execution

EDUCATION

B.Tech in Computer Engineering(pursing)	2025 – 2028
<i>CHARUSAT</i>	<i>CGPA: 6.68</i>
Diploma in Computer Engineering	2022 – 2025
<i>Government Polytechnic Kheda</i>	<i>CPI: 7.91</i>
SSC (Secondary School Certificate)	2022
<i>Best Secondary School</i>	<i>Percentage: 62.52%</i>